MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 7203

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal
See figure 1
Mounting position: Any
Weight: 4 ounces nominal (113.4 grams)

ABSOLUTE RATINGS: C Telegraphy

Parameter:	Fl	Ef	Eb	Ecl	Ec2	Ehk	Ib	Pgl	Pg2	Pp	T (seal and anode core)	tk	Cooling
Unit:	MHz	Vac	Vdc	Vdc	Vdc	Vdc	mAdc	W	W	W	°C	sec	
Maximum:	500	---	2,000	-250	300	±150	250	2	12	250	250	---	---
Minimum:	---	---											
		(note 11)	---	---	---	---	---	---	---	---	---	30	--- (note 2)
TEST CONDITIONS:	---	6.0	1,000	Adj	300	0	150	---	---	---	---	120	Note 3

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1					
1256	Electrode current (screen)	-		Ic2	-7.0	+3.0	mAdc
1261	Electrode voltage (grid)	-		Ec1	-32.0	-45.0	Vdc
1266	Total grid current	7	Eb = 2,000 Vdc; Ec1/Ib = 125 mAdc	Ic1	---	-15	µAdc
1266	Primary grid emission (control)	-	Ic1 = 70 mAdc; t = 120; anode and screen grid floating	Isg1	---	-250	µAdc
1266	Primary grid emission (screen)	-	Ec1 = 0; t = 120; Ic2 = 100 mAdc; anode floating	Isg2	---	-250	µAdc
1301	Heater current	-		If	2.30	2.90	Aac
1372	Current division (long pulse)	-	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk/ib = 1.0 a; prr = 11.0 ±1.0; tp = 4,500 µs (min)	egk ic1 ic2	8 --- ---	18 200 260	V ma ma
2212	Pulse emission	6	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk/ik = 1.5 a; Ef = 5.4 Vac; prr = 11.0 ±1.0; tp = 4,500 µs (min)	Δik	---	200	ma
	Quality conformance inspection, part 2						
1031	Low-frequency vibration	12	No voltages	---	---	---	---
1042	Shock, specified pulse	12	No voltages; 15 G (min); 11 ±2 ms; half sine wave	---	---	---	---
1331	Direct-interelectrode capacitance	-	EIA standard shields No. 320 and 321, or equivalent	Cgp Cin Cout	--- 14.2 4.0	0.06 17.2 5.0	pF pF pF
1336	Heater-cathode leakage	-	Ehk = +150 Vdc Ehk = -150 Vdc	Ihk Ihk	--- ---	150 150	µAdc µAdc
2214	RF useful power output	8	Class C amplifier; F = 470 to 500 MHz; Eb = 2,000 Vdc; Ec1 = -90 Vdc; Ec2 = 250 to 300 Vdc; Ic1 = 25 mAdc (max); Eg1/Ib = 250 mAdc; Ef = 5.5 Vac	Po	225	---	W
	Quality conformance inspection, part 3						
---	Life test (1)	-	Group C; rf useful power output	t	500	---	hrs
---	Life-test (1) end points:						
1266	Primary grid emission (control)	-	Ic1 = 70 mAdc; t = 120; anode and screen grids floating	Isg1	---	-250	µAdc
1266	Primary grid emission (screen)	-	Ec1 = 0; Ic2 = 100 mAdc; t = 120; anode floating	Isg2	---	-250	µAdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued						
---	Life-test (1) end points: -Continued						
1336	Heater-cathode leakage	-	Ehk = +150 Vdc Ehk = -150 Vdc	Ihk Ihk	--- ---	150 150	µAdc µAdc
2212	Pulse emission	6	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk/ik = 1.5 a; Ef = 6.0 Vac; prr = 11 ±1.0; tp = 4,500 µs (min)	Δik	---	100	mA
2214	RF useful power output	8	Class C amplifier; F = 470 to 500 MHz; Eb = 2,000 Vdc; Ec1 = -90 Vdc; Ec2 = 250 to 300 Vdc; Ic1 = 25 mAdc (max); Eg1/Ib = 250 mAdc; Ef = 5.5 Vac	Po	200	---	W
---	Life test (2)	-	Group C; Eb = Ec1 = Ec2 = 0; Ef = 6.6 Vac	t	500	---	hrs
---	Life-test (2) end points:						
1366	Interelement leakage resistance, cold	9	E = 100 Vdc; g1 negative E = 500 Vdc: g1 negative g2 positive g2 negative	Rg1k Rg1g2 Rg2k Rg2p	10 10 10 10	--- --- --- ---	Meg Meg Meg Meg
---	Life test (3)	10	RF useful output; airflow = 1.5 cfm	t	100	---	hrs
---	Life-test (3) end points:						
2214	RF useful power output	8	Class C amplifier; F = 470 to 500 MHz; Eb = 2,000 Vdc; Ec1 = -90 Vdc; Ec2 = 250 to 300 Vdc; Ic1 = 25 mAdc (max); Eg1/Ib = 250 mAdc; Ef = 5.5 Vac	Po	200	---	W
1011	Humidity	10,12	14 days duration; Ic1 = -20 µAdc (max) under post test conditions	---	---	---	---
1143	Forced cooling	4,10	Ec1/Ib = 250 mAdc	T(anode core)	---	225	°C
1155	Coolant-pressure drop versus coolant flow (forced air)	5,10	No voltages	---	---	0.35	In.H20

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NOTES:

1. The AQL for all tests listed under quality conformance inspection, part 1, shall be 0.65, inspection level II.

2. When the TUT is operated at 100 percent of maximum rated anode dissipation at an incoming air temperature of 25°C maximum, a minimum airflow of 3.8 cubic feet per minute (cfm) at sea level shall pass through the anode cooler. The static pressure drop across the TUT and socket (see Drawing 246-JAN) atan airflow of 3.8 cfm is approximately 0.30 inch of water. The pressure drop varies with the amount of escaping air and with the shape and construction of the air director. This rating applies at bias voltages less than 100 volts and frequencies less than 500 MHz. Air cooling on the TUT base shall be increased with increasing negative-grid bias or with increasing frequency, or a combination of both. In all cases of operation, a socket which provides forced-air cooling of the base shall be used and maximum seal temperature ratings must not be exceeded. The airflow shall be applied before or simultaneously with electrode voltages, and may be removed simultaneously with them.

3. The socket shown on Drawing 246-JAN shall be used in all electrical tests involving heater voltage. Forced-air cooling is permitted at the rate of 4.0 cfm maximum for the base and anode, unless otherwise specified in the specific test conditions. A separate source may be used for the base and anode, but neither shall exceed 4.0 cfm.

4. The forced cooling test shall be made as follows: At an ambient temperature of 25°C, both base and anode shall be cooled by applying an airflow of 3.8 cfm maximum at sea level from a single source using the infinite baffle system shown on figure 2, or equivalent. At the specified test conditions, the anode core temperature shall not exceed the specified limits. Temperature shall be measured by means of a thermocouple embedded in the top of the core, adjacent to the cooler. This shall be done by drilling a small hole, shallow enough so that the tube vacuum shall not be lost, placing the welded thermocouple junction therein, and then peening the edges of the hole to hold the thermocouple firmly in place. In all cases, good electrical continuity between the thermocouple and the metal area in close proximity shall be demonstrated before the cooling test can be performed.

5. An infinite baffle system (figure 2, or equivalent) with an airflow of 3.8 cfm at sea level shall be used. The static pressure drop is measured across the TUT and socket.

6. The input wave shape shall have a tr and tf of 25 µs maximum each, and the slope of the top of the pulse shall not be greater than 0.5 percent with a ripple not greater than 0.1 percent.

7. This test shall be the first test performed at the conclusion of the holding period.

8. The circuit and cavity shall be in accordance with Drawing 223-JAN.

9. This test shall be made 30 minutes after Ef is turned off. Rated airflow shall be maintained during this 30-minute interval. Measurement with Genreal Radio Megometer Model No. 1862C, or equivalent. Unused elements are to be left floating.

10. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

11. Maximum life may be obtained by adjusting the heater voltage in accordance with the application. A table of heater voltage versus frequency is presented as a guide for straight-through amplifier operation:

F (MHz)	Ef (Vac)
Up to 300	6.00
301 to 400	5.75
401 to 500	5.50

The heater voltages (nominal and derated) shall be maintained within ±5 percent (but not less than 5.4 Vac) when consistent operation and extended tube life are factors.

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on which will insure that suitable products ... is provided for obtaining information on the use of this specificati

NOTES: -Continued

12. At the conclusion of this test, the TUT shall satisfy the requirements of the total grid current test (method 1266) specified herein under quality conformance inspection, part 1.

13. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

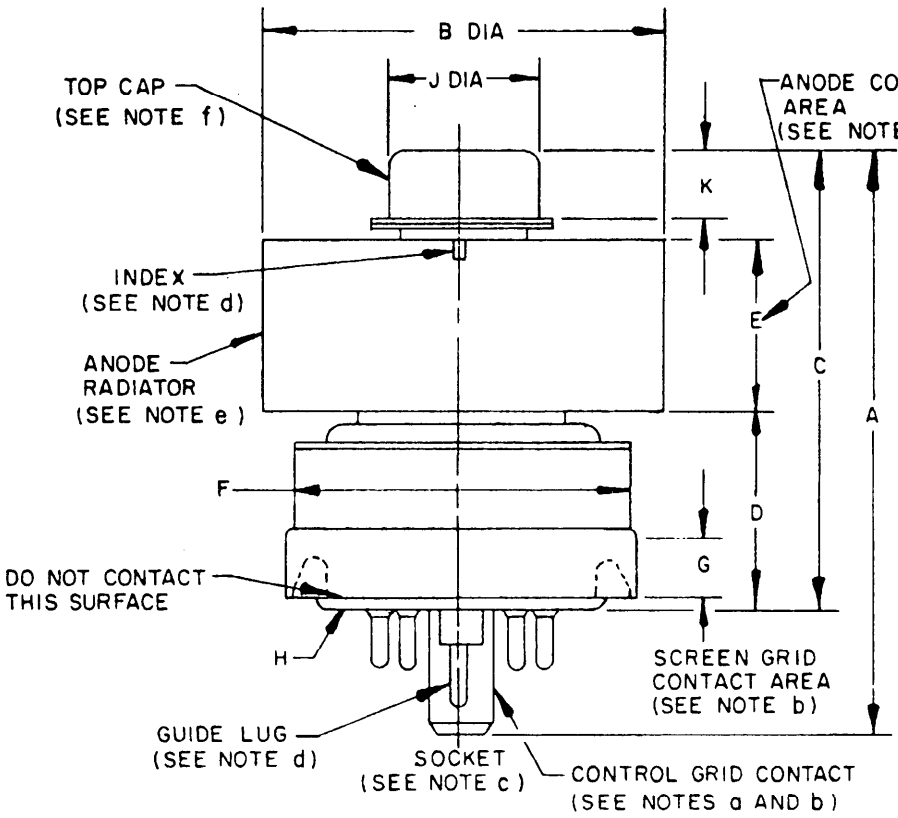
User activities:
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PIN CONNECTIONS	
Pin No.	Element
1	g2
2	k
3	h
4	k
5	int con
6	k
7	h
8	k
center post	g1
radiator & cap	a
base ring	g2

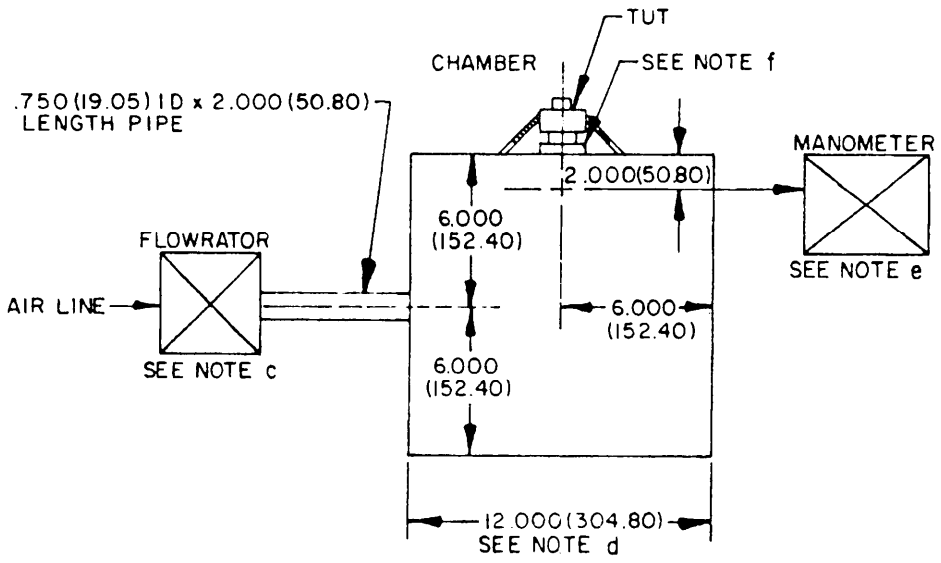
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	2.324 (59.03)	2.464 (62.59)
C	1.810 (45.97)	1.910 (48.51)
Quality conformance inspection, part 3 (SEE NOTE 10)		
B	1.610 (40.89)	1.640 (41.66)
D	.750 (19.05)	.810 (20.57)
E	.710 (18.03)	.790 (20.07)
F		1.406 (35.71)
G	.187 (4.75)	
H	Base: B8-236 (see note a)	
J	.559 (14.20)	.573 (14.55)
K	.240 (6.10)	

NOTES:

a. Pin alinement shall be checked by means of gage GB8-3. Dimensions of control-grid contact shall be inspected by means of gages specified on Drawing 246-JAN and shall be quality conformance inspection, part 2.
b. Alinement of anode, screen-grid, and control-grid contact surfaces shall be determined by means of gage specified on Drawing 168-JAN. Quality conformance inspection, part 2, shall apply.
c. Air-system socket shall be as specified on Drawing 246-JAN, EIMAC SK-600, or equal.
d. Location of guide lug of control-grid contact shall be referenced by a notch or arrow on the anode radiator in position shown.
e. Anode clamping shall be confined to anode radiator.
f. Top cap outline optional provided it meets requirements of dimensions J and K.

FIGURE 1. Outline drawing of electron tube type 7203.

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NOTES:
a. Dimensions are in inches.
b. Metric equivalents are in parentheses.
c. Fisher Porter flowrator model B4-27-10/77, or equal.
d. 12 inch (304.80 mm) cube inside dimensions, compound sealed.
e. F.W. Dwyer manometer, 0 to 1 inch (25.40 mm) of water (Fisher Scientific Company 11-295-5 draft gage), or equal.
f. Socket specified on drawing 246-JAN.

FIGURE 2. Baffle system.

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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